

Space-Air-Ground Integrated Network Resource Allocation Based on Service Function Chain

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Abstract—Future communication networks require higher bandwidth, greater coverage, and better throughput. The Space-Air-Ground Integrated Network (SAGIN) has the advantage of wide-area coverage and can cover global communications. It can meet the needs of network resources for maritime activities and remote mountainous areas, which is of great significance to the realization of a new generation of communications networks. With the development of the network and the emergence of delay-sensitive applications such as the Internet of Things, improving the delay performance of the system has received extensive attention. SAGIN involves multiple networks and is more complex than other networks. If there is no reasonable management between different networks, it is easy to lead to difficult link deployment and high time delay. On the basis of research and customization of SFC technology, it can provide a wide range of services and other advantages. It has significant performance in application scenarios involving multi network integration. In order to solve the above problems, this paper studies the SAGIN architecture of SFC based on business types. A service function chain mapping method based on delays prediction is proposed. Calculate the delay of the deployment path and select the path with the lowest delay as the SFC mapping path. The service model is constructed according to the mapping path, and the network slices are divided based on the service type. The simulation results show that the SFC mapping algorithm based on time delay prediction is compared with the traditional SFC mapping scheme. The algorithm does not affect other indicators,

the CPU resource utilization rate is 27.8% higher, and the link resource utilization rate is 22.7% higher. The service acceptance rate increased by 21.5%, the latency performance increased by 38.2%, and the total resource consumption is reduced by 25.2%.

Index Terms—Space-air-ground integrated network, service function chain, delay sensitive, resource allocation.

I. INTRODUCTION

INTERNET of things (IoT) [4] can provide connections for large-scale devices and sensors. It is considered as an enabling technology for building intelligent homes, intelligent transportation, intelligent cities and intelligent earth [35]. It has become an important technology to bring cyberspace into human life in the 21st century [8]. With the development of technologies such as the IoT, cloud computing, and artificial intelligence, traditional terrestrial wireless communications have exponentially increased in terms of the number of users and supported services [3], and future networks will require more resources. In addition, remote areas and high-altitude areas, especially areas with harsh environments such as oceans and mountains, have the problem of ground cellular or optical fiber network coverage loopholes due to their geographical restrictions [13]. Traditional terrestrial communications can no longer meet a large number of communications needs, and new network architectures must be developed to adapt to different scenarios and applications. After decades of development, the development of global space infrastructure has established a solid foundation, and a complete and supporting space-based information system has been basically completed. Satellite application technologies and industries have also matured. However, there is a phenomenon of fewer applications on the ground and satellite resources have not been fully utilized [24].

The current Internet architecture cannot meet the growing demand of end users, and the current Internet architecture is rigid [9]. The integrated network including space-based network, air-based network and ground-based network has attracted extensive attention [2]. SAGIN has the advantages of flexible networking, wide coverage and unlimited by geographical conditions. It is applicable to many fields, especially combined with satellite networks to achieve full coverage of oceans and mountains [16]. In SAGIN, different application scenarios correspond to different business flow processing to form a SFC. The introduction of SFC realizes the elastic service of business flow. Since the concept of SAGIN was put forward, researchers have done extensive research, but most of them are limited to network integration and access, ignoring the possible problems in practical application. In practical applications, customers

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have higher and higher requirements for QoS. Delay, as a key factor affecting user experience, is closely related to QoS [26]. In practical application fields, medical, military and artificial intelligence all have high requirements for time delay. In the military field, military experts predict that the war in the 21st century will be an information war. The lower the time delay, the more information can be obtained, the better the grasp of the war situation, and the greater the chance of winning. In the medical field, telemedicine can provide help to the lack of medical resources or special environment. In the field of medical rescue, time is life. Taking 4 K technology as an example, the required network delay needs to be between 12 ms and 17 ms. The ultra-low-latency communication scenario requires the end-to-end delay of the service to be 3 ms to 5 ms. The delay at the end is less than 3 ms.

By analyzing the current research status, the challenges are as follows:

- 1) SAGIN network includes space-based network, air-based network and ground-based network. The deployment process of inter network links is complex;
- 2) With the rapid growth of delay-sensitive applications, networks demand to guarantee low latency to ensure network service quality. The configuration of network resources directly affects the network latency [12]. In practical application, there is the problem of sharing multiple SFC bandwidth and link resources, resulting in unreasonable total resource allocation;
- 3) The network load is highly dynamic, and the deployed link has the phenomenon of delay jitter. It is complex to determine the delay of SFC.

In order to solve the above problems, this paper proposes a resource allocation algorithm based on SFC. The main contribution of this paper can be concluded as follows:

- 1) Aiming at the difficulties of unified management of network equipment and link deployment in SAGIN system [20], Network Function Virtualization (NFV) technology and SFC are used to solving the problems of equipment and link in multi network integration process. In the link mapping stage, k -shortest path algorithm is used for link mapping to make the link logic simpler. In addition to determining the shortest path, it also determines the second shortest path and the third shortest path up to the k -th shortest path;
- 2) In the SAGIN system, business types are classified, and similar businesses form a unified business framework. Deploy SFC according to service type. NFV technology is used to sense the load of network nodes and links to ensure the maximum utilization of network resources;
- 3) Set the delay limit according to the historical load of each type of service. By detecting the load of the network to dynamically expand or reduce the delay boundary, and finally determine the delay of SFC.

The remainder of this paper is organized as follows. In Section II, the existing research on SAGIN is reviewed. Section III describes the network model and deployment path reliability. In Section IV, we introduce the steps of the algorithm. In Section V, we describe in detail our proposed delay sensitive service function chain mapping algorithm. Section VI evaluates the performance of our method and other methods. Section VII summarizes this paper.

II. RELATED WORK

In this section, we will review the existing literature on service chain based algorithms in SAGIN. According to the logical classification of the algorithm, it can be divided into optimization algorithm and heuristic algorithm.

A. Optimization Algorithm

SAGIN network involves three network segments. Due to the heterogeneity of different levels, effective resource allocation is needed. A graph based joint optimization algorithm for resource block and power allocation is proposed in literature [30]. At the same time, combined with the power allocation method of multi-stage water injection, the joint optimal solution can be obtained after finite iterations by using the alternating descent method. The algorithm has significant improvement in resource allocation rate and fairness. The application of the Internet of things requires low delay and high bandwidth. SAGIN with multi-function is widely used in the Internet of things. In literature [31], the author studies the task unloading of the Internet of things in the SAGIN scenario, proposes a computational unloading strategy, uses unmanned aerial vehicle (UAV) to fly along a given trajectory, collects the tasks of the Internet of things devices in the coverage area, and then makes an online unloading decision. The author in literature [32] studies the delay oriented computing task scheduling problem of IoT in SAGIN, and designs a scheduling strategy to minimize the unloading and computing delay of all tasks when the energy capacity of UAV is limited. A new deep risk sensitive reinforcement learning algorithm is proposed. The algorithm evaluates the risk of each state, measures the energy consumption beyond the constraint, and searches for the optimal parameters of weighted delay and risk minimization when learning the optimal strategy. The algorithm effectively reduces the task processing delay under the condition of meeting the UAV energy capacity constraints. However, the processing process is more complicated and the cost is higher.

B. Heuristic Algorithm

How to improve resource utilization and service quality by improving the SFC orchestration and mapping scheme is an important research content in the NFV-based world-ground integrated network. Literature [14] describes two deployment modes of SFC in satellite communication networks: multi-domain mode and satellite formation mode, and proposes two algorithms suitable for different service request to deploy service function paths, which effectively reduces the delay and packet loss rate, but it lacks a unified network model to solve the SFC deployment problem. Literature [28] proposed a reconfigurable service provision architecture based on the service function chain. Through a heuristic greedy algorithm, the different characteristics of air and ground nodes are used to balance resource consumption, saving a lot of communication resources, but it cannot change according to network status, optimize the SFC migration process. In order to meet the computing intensive Internet of things applications and different QoS requirements, the author of literature [38] proposed a SAGIN mobile edge network to realize mobile edge computing. A scheduling scheme

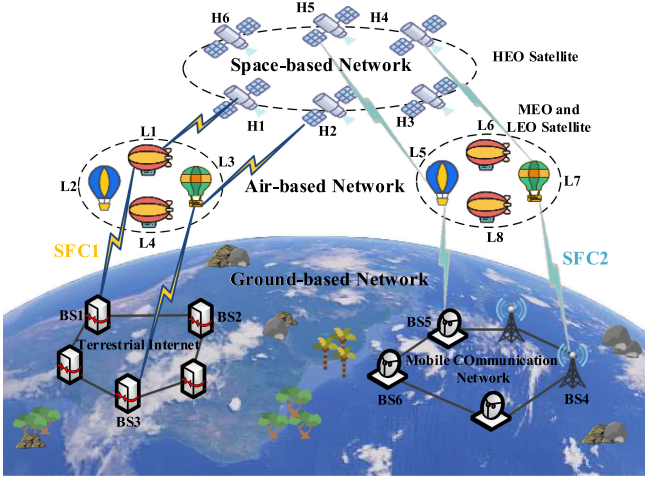


Fig. 1. Space-air-ground integrated network model based on service function chain.

based on reinforcement learning is proposed in literature [2] to deal with the problem that it is difficult to determine the channel conditions in remote areas. It effectively solves the link resource allocation, but does not fully consider the different business needs of users.

All the above studies have achieved certain experimental results. We have summarized the experience in the above research and made innovations based on actual application scenarios. First of all, we have designed algorithms to solve these problems from the perspective of SFC based on SAGIN's resource allocation and protection of users' business needs. Secondly, it focuses on the time delays problem of SFC in SAGIN, and uses the SFC mapping algorithm based on time delay prediction to improve, which effectively improves the experimental performance. The existing SAGIN resource allocation algorithm does not pay attention to the user's QoS, and integrates the allocation of network resources. Therefore, our work is innovative.

III. SPACE-AIR-GROUND INTEGRATED NETWORK MODE

The SAGIN architecture proposed in this paper is based on SFC and classified according to business classification. Different businesses are divided into different SFCs. Fig. 1 shows the SAGIN model based on SFC. The model is divided into three layers: space-based network layer, air-based network layer and ground-based network layer. Terrestrial internet and mobile communication network are terrestrial base stations accessed by different users using different wireless access methods. A high elliptical orbit (HEO) satellite can cover multiple medium earth orbit (MEO) and low earth orbit (LEO) satellites. HEO satellite is in the space-based network layer. MEO and LEO satellites are in the air-based network layer. The satellites shown in the Fig. 1 have VNF [7]. Satellites in the ground-based network layer are responsible for user access, satellites in the space-based network layer are responsible for data transmission, and satellites in the space-based network layer are responsible for data processing. Fig. 1 contains two SFCs that can serve two different business types.

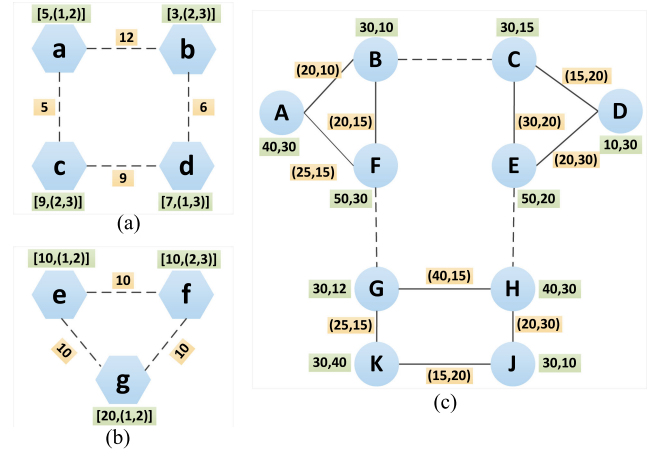


Fig. 2. The diagram of virtual network and substrate network.

The SFC1 traffic flows from the ground network node BS3 to the L3 node of the air-based network layer through the link, and then to the ground BS1 node through satellites H2, H1 and L1 in turn. SFC2 traffic flows from mobile communication network node BS5 to L5 node in air-based network layer through link, and then to BS4 node in mobile communication network through satellites H5, H4 and L7 in turn. The data is transmitted by the node equipment of the route and processed in sequence [10].

A. Physical Network Model

In general, we use a weighted undirected graph $G_V = \{N_V, L_V\}$ to represent a virtual network, therein [15], $N_V = (n_V^i, i = 1, 2, 3, \dots)$ represents the set of all virtual nodes, the CPU resource demand of each virtual node can be expressed as $CPU(n_V^i)$, and the candidate domains of each virtual node are represented by $CD(n_V^i)$, $L_V = \{l_V^k, l_V^k = (n_V^i, n_V^j), k = 1, 2, 3, \dots\}$ represents the set of all virtual links, $BW(l_V^k)$ is used to represent the bandwidth resource demand of virtual link l_V^k . An example of a virtual network is shown in Fig. 2(a)(b). There are two virtual networks in the Fig. 2(a)(b). The virtual node is represented by a hexagon. The CPU resource requirement of each virtual node is represented by the first number in the rectangle. The number in the square brackets represents the candidate domain of the virtual node. Each virtual node can have two candidate domains. The virtual link is represented by a dotted line, and each virtual link has its corresponding bandwidth resource requirement, which is represented by the number next to the dotted line.

The substrate network can also be represented by a weighted undirected graph $G_S = \{N_S, L_S\}$, therein, $N_S = (n_S^i, i = 1, 2, 3, \dots)$ represents the set of all physical nodes, the available CPU capacity of physical nodes are represented by $CPU(n_S^i)$, and the delay of each physical node is represented by $delay(n_S^i)$, $L_S = \{l_S^k, l_S^k = (n_S^i, n_S^j), k = 1, 2, 3, \dots\}$ represents the set of all physical links, $BW(l_S^k)$ is used to represent the bandwidth capacity of physical link l_S^k , the delay of each physical link is represented by $delay(l_S^k)$. The substrate network is shown in Fig. 2(c). The substrate network is divided into three different domains, and the physical nodes are represented by circles. Each physical node has its available CPU capacity, represented by the first number in the rectangle next to the node, and each physical

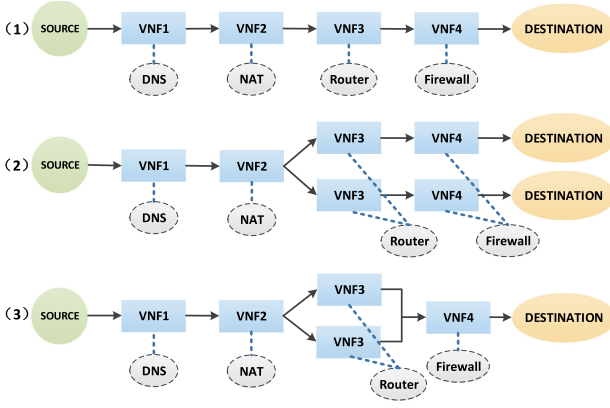


Fig. 3. The forms of SFC requests.

node has its delay, represented by the second number in the rectangle next to the node. At the same time, the intra-domain links of the substrate network are represented by solid lines and the inter-domain links are represented by dashed lines. The brackets next to the physical link have two parameters. The first represents the available bandwidth capacity of the physical link, and the second represents the delay of the physical link.

B. Business Type Classification

There are similar business scenarios in SAGIN [19] system, such as those with high bandwidth requirements and those with high delay requirements. According to the request scale of users, the requirements of link resources and delay, the whole application service is divided into six categories. They are: large-scale and high bandwidth, large-scale and low delay, low-scale and high bandwidth, low-scale and low delay, high bandwidth and low delay, low bandwidth and low delay. SFCs are matched according to the service type [23]. Each SFC includes ground gateway, LEO and MEO satellite, HEO satellite and corresponding links. If the user requests large-scale and requires high bandwidth, the corresponding service type is large-scale and high bandwidth. If the required bandwidth is large and the delay requirement is low, the corresponding service type is high bandwidth and low delay. If the scale of the user's request is very small, but the delay requirement is high, the corresponding service type is low-scale and high delay.

C. SFC Request Modeling

Each SFC request shall include the nodes, links connecting nodes, VNF sequence required by service flow, bandwidth requirements, delay requirements, etc. Therefore, the SFC request model can be represented by $Q = \{N, L, V, B, D\}$. Where N represents the node, L represents the link, V represents the VNF sequence, B represents the bandwidth requirements of the SFC request, and D represents the delay requirements of the SFC request. According to the literature, there are three common types of SFC requests, as shown in Fig. 3. The first type is that the origin corresponds to the destination node one by one, and there is only one link between the origin and the destination node. The second type is that the relationship between the origin and the destination node is one to many. Starting from the origin, the link of the path is different and the destination node is also

different. In the third type, the source node and destination node are one-to-one, but there are multiple links from the source node to the specified target node.

In addition, NM_i^u is used to represent the mapping of nodes in the SFC request. NM_i^u equal to 1 means that n_u is mapped to n_i , and equal to 0 means that it is not mapped to. $LM_{i,j}^{m,n}$ indicates the link mapping, $LM_{i,j}^{m,n}$ equals 1 indicates that $l_{m,n}$ is mapped to $l_{i,j}$, and equal to 0 means that it is not mapped to.

IV. TARGET OF DEPLOYMENT PATH

According to the request of SFC, the optimal deployment path in the current network state is selected from multiple paths by evaluating load conditions, normal operation probability, and various path delays. TD_q is used to represent the delay of the deployment path. The delay is determined by many factors, including node mapping delay, link mapping delay, data transmission delay, data processing delay and so on.

The network node normal operation probability $NO_i(q)$ and link connectivity probability $LO_{i,j}(q)$ are evenly distributed on $[e^{-0.01(q-1)}, e^{-0.01q}]$, where q is the total number of SFCs in the network services.

The node load parameter is the ratio of the computing resources used on the node and the node computing resources required by the SFC request to be deployed in the network to all the computing resources of the node, expressed as:

$$NL_i^n = \frac{C_i^j + \sum_{u \in V} NO_i \times R_i^n}{C_i^m}. \quad (1)$$

The link load parameter is the ratio of the occupied bandwidth on the link and the link bandwidth required in the SFC request to the total bandwidth of the link, expressed as:

$$LL_{i,j}^l = \frac{C_{i,j}^{m,n} + \sum_{e_{m,n} \in L} LO_{i,j} \times R_{i,j}^l}{C_{i,j}^m}. \quad (2)$$

The total resource consumption of the deployment link is expressed as follows, where C_i represents the resource capacity on the computing node and $C_{i,j}^{m,n}$ represents the total bandwidth resources on the link.

$$TC_q = \sum_{i \in N} C_i^j + \sum_{l_{i,j} \in L} C_{i,j}^{m,n}. \quad (3)$$

The total service delay of SFC request deployment to the network consists of data processing delay and deployment link delay, expressed as:

$$TD_q = \sum_{v_m \in N} D_m^T + \sum_{e_{i,j} \in L} D_{i,j}^T. \quad (4)$$

In order to find the optimal path in multiple paths, the objective function is set to minimize the deployment path delay, expressed as:

$$\min TD_q. \quad (5)$$

The upper limit of the total delay of SFC deployment path is the delay specified by the user:

$$TD_q \leq UD_q. \quad (6)$$

TABLE I
SYMBOL SUMMARY

Symbol	Description
H1-H6	LEO satellites
L1-L8	LEO and MEO satellites
NM_i^U	mapping of nodes n_u to n_i
$LM_{i,j}^{m,n}$	mapping of links $l_{m,n}$ to $l_{i,j}$
NO_i	probability of normal operation of node n_i
$LO_{i,j}$	probability of normal operation of link $l_{i,j}$
NL_i^n	node load parameter
$LL_{i,j}^l$	link load parameter
TC_q	total resource consumption of deployment link
C_i^j	amount of resources used by the node
$C_{i,j}^{m,n}$	amount of resources used by the link
R_i^n	the amount of resources required to map to a node
$R_{i,j}^l$	the amount of resources required to map to a node
C_i^m	total resources owned by node
$C_{i,j}^m$	total resources owned by link
TD_q	total service delay
D_m^T	data processing delay
$D_{i,j}^T$	deployment link delay
UD_q	user specified delay
P_q	deployment path

The amount of resources required by the node is less than or equal to the amount of resources owned by the node:

$$NL_i^n \leq 1. \quad (7)$$

The amount of resources required by the link is less than or equal to the amount of resources owned by the link:

$$LL_{i,j}^l \leq 1. \quad (8)$$

We summarize all the symbols in Table I.

V. SERVICE FUNCTION CHAIN MAPPING ALGORITHM BASED ON DELAY SENSITIVITY

There will be multiple feasible paths in the SFC request mapping to the network. In order to make the deployment path have lower delay and minimum resource overhead, this paper proposes a service function chain mapping algorithm based on delay awareness [33]. Input SFC request Q and physical network G_v , and output as deployment link delay TD_q and deployment path P_q . The main work of the algorithm is to classify SFC requests according to the size of the request, bandwidth requirements and delay requirements. Each request belongs to a business type. In each service, KSP algorithm is used to select k paths with the shortest delay and less resource overhead. Calculate the resource overhead and delay of each

Algorithm 1: Service Function Chain Mapping Method Based on Delay Sensitivity.

Input: SFC request Q , physical network G_v ;
Output: Deployment path P_q , total service delay of SFC request deployment to the network TD_q ;

- 1: Initialization $P_q = 0, TD_q = 0$;
- 2: Determine the service type corresponding to SFC according to the request scale, bandwidth and delay;
- 3: Generate VNF sequence F according to SFC request;
- 4: KSP algorithm is used to select the first k shortest paths in the corresponding service types;
- 5: **for** $k=1:K$ **do**
- 6: **if** $NO_i(q) > 0.5$ and $NL_i^n \leq 1, LO_{i,j}(q) > 0.5$ and $LL_{i,j}^l \leq 1$ **then**
- 7: calculate the data processing delay D_m^T and deployment link delay $D_{i,j}^T$ of path P_k ;
- 8: **else**
- 9: $D_m^T = -1, D_{i,j}^T = -1$;
- 10: **end if**
- 11: calculate the total delay TD_k of path P_k ;
- 12: **end for**
- 13: select the path with the least total delay as the deployment path P_q ;
- 14: **if** $TD_q \leq UD_q$ **then**
- 15: instantiate SFC and update node and link load;
- 16: return P_q, TD_q ;
- 17: **else**
- 18: SFC cannot be deployed in the physical network;
- 19: **end if**

path, and select the path with the least delay as the deployment path of SFC. If the delay of the deployment path is less than the delay specified by the user, a SFC is formed, otherwise the SFC deployment fails. The specific process of the algorithm is shown in **Algorithm 1**.

VI. SIMULATION EXPERIMENT AND ANALYSIS

In this part, we describe the setting of the simulation environment and give the simulation results. We evaluate the difference between our algorithm and other algorithms through five evaluation criteria, including CPU resource utilization, link resource utilization, SFC request acceptance rate, deployment path delay and resource consumption rate. The algorithms compared in this section are SPO algorithm [7], CLF algorithm [17], and DMRT-SL algorithm [5].

A. Experimental Environment Setting

Our experiment is implemented on IDEA using Java. We use GT-ITM [1] to generate the topology of SFC requests.

B. Simulation Parameter Setting

The slice network topology constructed in the simulation experiment is shown in Fig. 4. The topology is composed of 8 ground gateway nodes, 8 MEO and LEO satellite nodes, 4 HEO satellite nodes, ground links, satellite ground links and inter satellite links. In the simulation, the required bandwidth

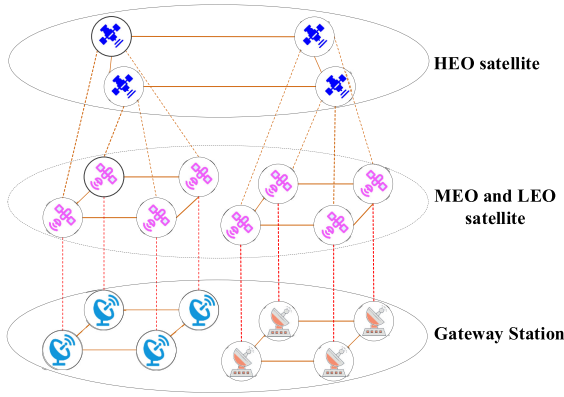


Fig. 4. Network slice topology.

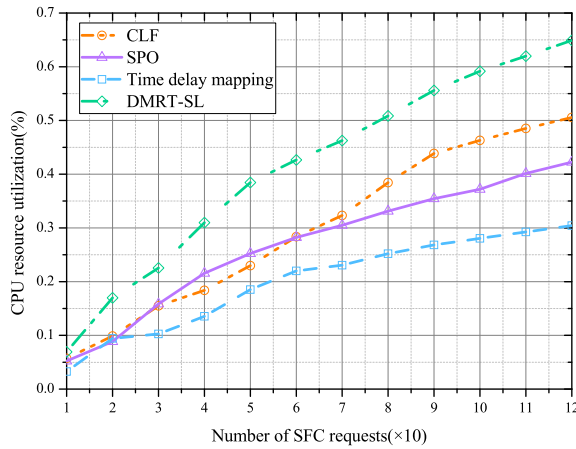


Fig. 5. CPU resource utilization comparison.

and delay of the six service types are different. The resources of large-scale high bandwidth, large-scale low delay, low-scale high bandwidth, low-scale low delay, high bandwidth low delay, low bandwidth low delay are set to 600, 400, 500, 400, 500, 300 respectively. The SFC request type generates a continuous arrival network with the same probability, and the bandwidth required in the request is evenly distributed at [10,100] MHz. The delay in the request is distributed in [0,200] ms, and the upper limit of delay specified by the user is set to a fixed value of 200 ms.

C. Experimental Results and Analysis

In order to verify the performance of the SFC algorithm based on delay perception proposed in this paper, we designed five simulation experiments [29].

Experiment 1: CPU resource utilization comparison. In order to compare the CPU resource utilization of the algorithm proposed in this paper with that of other algorithms, we conducted a CPU resource utilization comparison experiment.

Experimental results analysis: The experiment compares the CPU resource utilization of this algorithm, SPO algorithm, CLF algorithm and DMRT-SL algorithm under different SFC requests. As can be seen from Fig. 5, the CPU utilization increases gradually with the increase of the number of SFC request, and the increasing range is gradually slow. The reason is that in the early stage of operation, CPU capacity resources are relatively

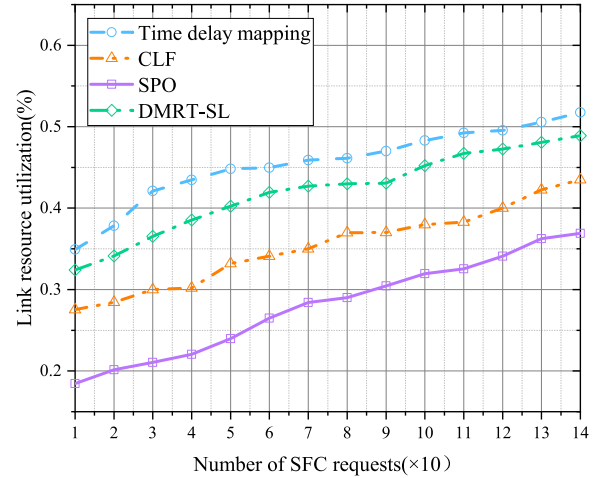


Fig. 6. Link resource utilization comparison.

rich. Fig. 5 shows that the trend of CPU resource utilization of DMRT-SL algorithm is close to that of CLF algorithm. The utilization rate of these two algorithms is higher than that of SPO algorithm and the algorithm proposed in this paper. The reason is that DMRT-SL and CLF algorithms do not fully consider the occupation of resources. The CPU resource utilization of the algorithm proposed to this paper is the lowest, because the amount of CPU resources is considered in the algorithm design, so that the deployment path can reduce the consumption of CPU resources as much as possible. As shown in Fig. 5, compared with other algorithms, the performance of the algorithm proposed to this paper is remarkable.

Experiment 2: link resource utilization. In order to compare the link resource utilization between the proposed algorithm and other algorithms, we conducted a link resource utilization comparison experiment.

Experimental results analysis: The experiment compares the link resource utilization under different SFC requests. It is clear from Fig. 6 that the link resource utilization increases with the number of SFC requests. Fig. 6 shows that this algorithm has the highest link resource utilization. Because the algorithm proposed in this paper divides the service flow according to the SFC requests type, ensures the effective utilization of resources, reduces the waste of resources, and maximizes the link utilization rate. When the SFC request is greater than 120, the link resource utilization rate increases by 13.2%. The link utilization rate of the SPO algorithm is low, because the SPO algorithm adopts the method of combining and splitting requests to deploy links, resulting in excessive consumption of link resources.

Experiment 3: deployment path delay. In order to compare the mapping path delay between this algorithm and other algorithms, we carried out the comparison experiment of shooting path delay.

Experimental results analysis: The experiment compares the deployment path delay under different SFC requests. It can be clearly seen from Fig. 7 that the deployment path delay shows an upward trend as the number of SFC requests increases. Fig. 7 shows that the mapping path delay of the proposed algorithm is the lowest because the algorithm is path mapping based on delay prediction, and the algorithm selects the deployment path

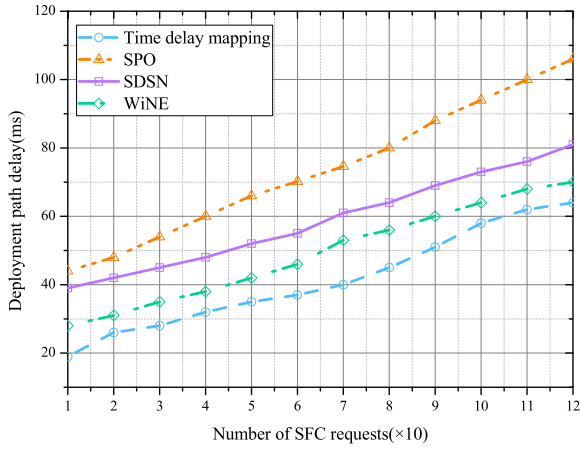


Fig. 7. Deployment path delay comparison.

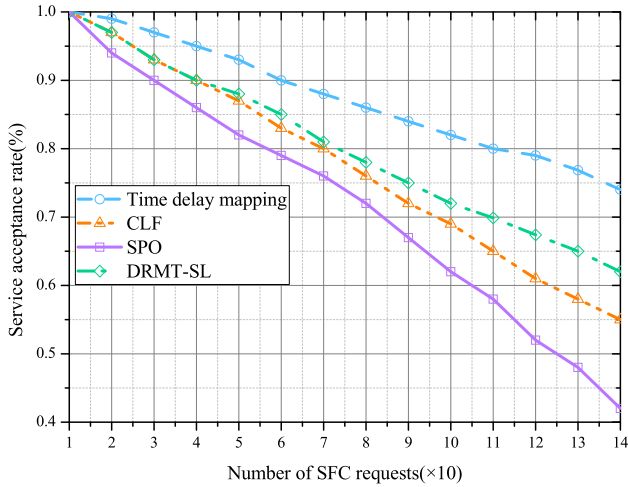


Fig. 8. Comparison of service acceptance rate.

according to the delay of the path. With the increase of SFC requests, the advantage of this algorithm is more obvious.

Experiment 4: service acceptance rate. In order to compare the service acceptance rate of this algorithm with other algorithms, we carried out a comparative experiment of service acceptance rate.

Experimental results analysis: The experiment compares the service acceptance rate of different SFC requests. It can be clearly seen from Fig. 8 that the service acceptance rate is the highest in the early stage of operation. As the number of SFC requests increases, the service acceptance rates decreases. Because when the use of network resources reaches a certain range, the network are close to saturation and the available resources are reduced, so new network requests cannot be accepted. It can be seen from Fig. 8 that the service acceptance rate of the algorithm decreases slowly, and the overall performance is better than that of the SPO, CLF and DMRT-SL algorithms. Because the algorithm reduces the consumption of network resources as much as possible, thereby improving the service acceptance rate.

Experiment 5: resource consumption rate. In order to compare the total resource consumption rate of this algorithm with other algorithms, we carried out a comparative experiment of resource consumption rate.

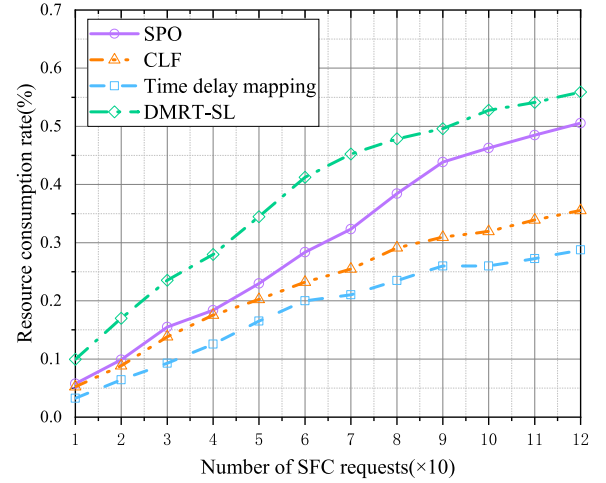


Fig. 9. Comparison of resource consumption rate.

Experimental results analysis: The experiment compares the resource consumption rate under different SFC requests. It can be clearly seen from Fig. 9 that as the number of SFC requests increases, the resource consumption rate shows an upward trend. As the number of SFC requests increases, so does the amount of resources required. The resource consumption rate of the SPO algorithm and the DMRT-SL algorithm is much higher than the other two algorithms, because the SPO and DMRT-SL algorithms do not fully consider the resource consumption factor. The total resource consumption rate of the algorithm proposed in this paper is the lowest. With the increase of the number of SFC requests, the advantages of the algorithm in this paper are more significant.

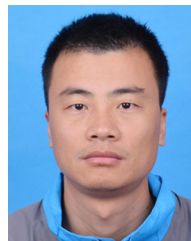
VII. CONCLUSION

The research on SAGIN resource allocation is a hot issue in the field of space-based communication. This paper mainly studies the resource allocation problem based on SFC in SAGIN network, and proposes a delay sensitive SFC mapping algorithm for delay sensitive applications. Simulation results show that the algorithm has significant advantages in improving link resource utilization, reducing delay and improving service acceptance rate. Compared with other SFC algorithms, the link resource utilization is increased by 22.7%, the service acceptance rate is increased by 21.5%, the delay is reduced by 38.2%, and the total resource consumption is reduced by 25.2%. Therefore, the scheme proposed in this paper has great research significance.

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